Pressure

Pressure (P) is defined as the force acting normally to an unit area on a surface.

$$\text{Pressure (P)} = \frac{Force\ (F)}{Area\ (A)}$$

- 1 atm = 100 kPa
- Gas pressure can be measured by **Bourdon guage**

Unit: Pa or Nm^{-2}

1. Calculate the pressure at the point of contact when a 70 kg person

(a) stands on the floor (total area of his feet = 4.0×10^{-2} m^2)

(b) stands on one nail (area of nail tip = 4.0×10^{-6} m^2)

(c) lies on 1000 nails

(a) 17167.5 Pa

(b) 171675 kPa

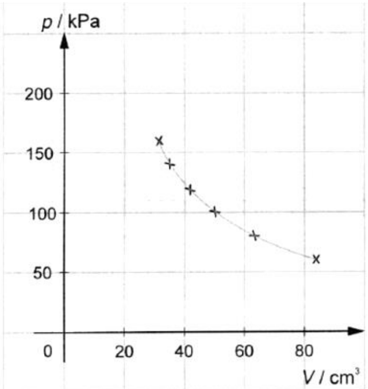
(c) 171675 Pa

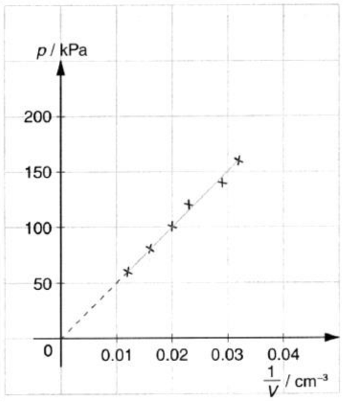
Boyle's law

The pressure of a fixed mass of gas at a constant temperature is **inversely proportional** to its volume.

$$p\ \alpha\ \frac{1}{V} \quad \text{(constant T)}$$

$pV = constant$ or $p_1V_1 = p_2V_2$ (constant T)





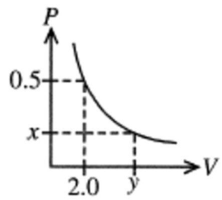
2. A syringe contains 50 cm^3 of air at atmosphere pressure (100 kPa). If the volume is reduced to 30 cm^3, what will be the new pressure? Assume that the temperature remains constant.

$$P_1V_1 = P_2V_2$$
$$(100\ 000)(50) = P_2(30)$$
$$P_2 = 166\ 666.7\ Pa$$

3. The opening of the hand pump shown above is sealed. The initial pressure of the air inside the hand pump is 1 atm. When the handle of the hand pump is pushed slowly, the length of the air inside it drops from 30 cm^3 to 24 cm^3. Find the final pressure of the air inside the hand pump

As the air is compressed slowly, its temperature remains unchanged.
By Boyle's law, PV = constant
$1.0 \times 30 = P \times 24$
$P = 1.25$ atm
The answer is C.

4. The pressure-volume (P-V) graph of an ideal gas at a fixed temperature is shown above. Find the value of the product xy.



A. 0.25
B. 0.5
C. 1.0
D. 2.0

By Boyle's law, PV = constant,
$0.5 \times 2.0 = xy$
$xy = 1.0$
The answer is C.

5. The pressure of a fixed mass of gas is increased from 100 kPa to 200 kPa while its temperature is kept constant. What is the percentage change in its volume?

A. -50%
B. -25%
C. 50%
D. 100%

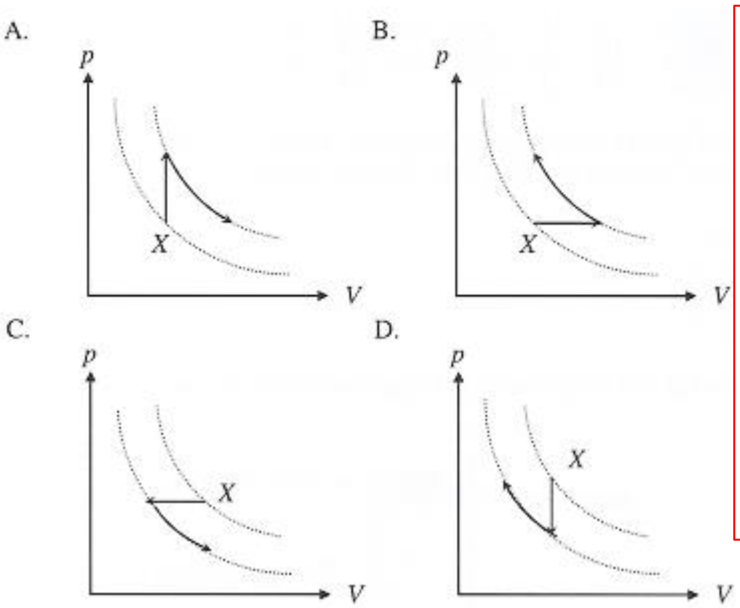
A

6. A fixed mass of ideal gas is pressurized under a constant temperature. Which of the following descriptions about the gas molecules is correct?

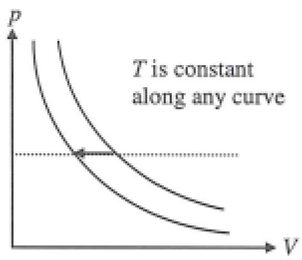
	Average K.E	Average seperation
A.	unchanged	decreases
B.	unchanged	increases
C.	increases	unchanged
D.	increases	decreases

As the temperature is kept constant, the average kinetic energy of the gas molecules remains unchanged. Options C and D are incorrect.
By Boyles' law, PV = constant.
As P increases, V decreases. The average separation between gas molecules decreases.
The answer is A.

7. X represents the state of a fixed mass of gas. The temperature of the gas is first decreased, then the pressure of the gas is decreased at constant temperature. Which of the diagram is a possible path?



C



Along any p-V curve, the temperature is constant. Keeping p constant, the decrease in V is accompanied by a decrease of temperature. Thus, the curve near to the origin is of lower temperature.
When pressure is decreased at constant temperature, X will move along the p-V curve, in downward direction (as p decreases).

C

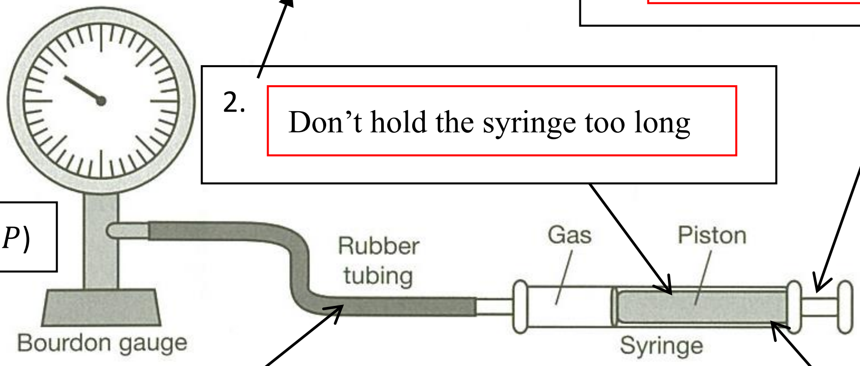
Bourdon gauge

To | Prevent temperature change

1. Change the gas volume slowly

2. Don't hold the syringe too long

Measure gas pressure (P)



Measure gas volume (V)

3. Use a short rubber tubing

4. Use a longer syringe

To | Minimize errors in measuring gas volume

8 The above apparatus is used to study the relation between the pressure and volume of a fixed mass of gas at constant temperature. Which of the following can improve the accuracy of the experiment?

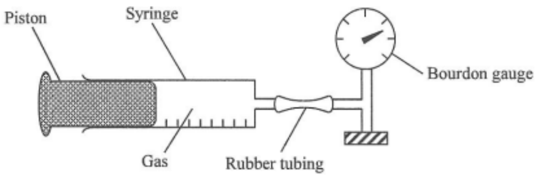
A

A. using the larger syringe

B. pushing the piston quickly

C. using a longer length of rubber tubing

D. setting a control experiment with the bourdon guage removed



9 The figure below shows the set-up used to study the relationship between the volume and pressure of a fixed mass of gas at constant temperature. A student pushes the piston and takes several sets of readings of the pressure and volume of the gas inside the syringe.

Which of the following statements about the experiment is/are correct?

(1) If the volume is plotted against the reciprocal of the pressure, a straight line passing through the origin will be obtained.

(2) By pushing the piston continuously, the pressure will increase but not exceed 100 kPa, which is the atmospheric pressure.

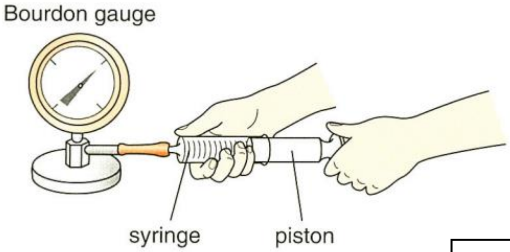
(3) To increase the accuracy of the experiment, the student should push the piston slowly.

A. (3) only

B. (1) and (2) only

C. (1) and (3) only

D. All of the above.



C The gas is expected to obey Boyle's law:

pV = constant $\Rightarrow$ V = constant $\times \dfrac{1}{p}$

∴ (1) is correct.

The pressure will increase and can exceed 100 kPa when the volume of the gas keeps decreasing.

∴ (2) is incorrect.

Pushing the piston slowly is a precaution for the experiment to prevent any change in temperature inside the syringe.

∴ (3) is correct.

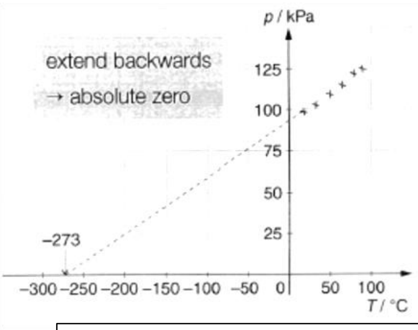
C

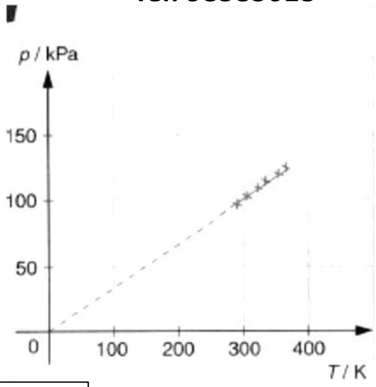
必究

Pressure law

For a gas with a fixed mass (m) and volume (V), its pressure (P) is **directly proportional** to its **Kelvin temperature** (T) Unit: (K)

$$P \propto T \quad \text{(constant V)}$$





$$\frac{p}{T} = \text{constant or } \frac{P_1}{T_1} = \frac{P_2}{T_2} \text{ (constant V)}$$

For example: 30 ºC = ______K

___ºC = 380 K

10. The pressure of a car tyre is 300 kPa at 27 ºC. After a journey the temperature of air in the tyre rises to 57 ºC. Assuming that the volume of the tyre has remained constant, find the new pressure of the tyre.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{300000}{27+273} = \frac{P_2}{57+273}$$

$$\rho_2 = 330000\ Pa$$

11. A sealed vessel of fixed volume contains air at atmospheric pressure and at a temperature of 27°C. If the pressure of the air has to be increased to 200000 Pa, what should the temperature of the air be increased to?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{100000}{27+273} = \frac{200000}{T_2}$$

$$T_2 = 600K$$

12. The maximum gas pressure that can be tolerated by a hair spray container is 3 atmosphere. The gas in the spray container at 32ºC has pressure of 2.7 atmosphere. At which temperature will the spray container explode?

Since the volume is constant,

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{2.7}{273+32} = \frac{3}{273+x}$$

$$x = 65.9 \approx 66\text{°C}$$

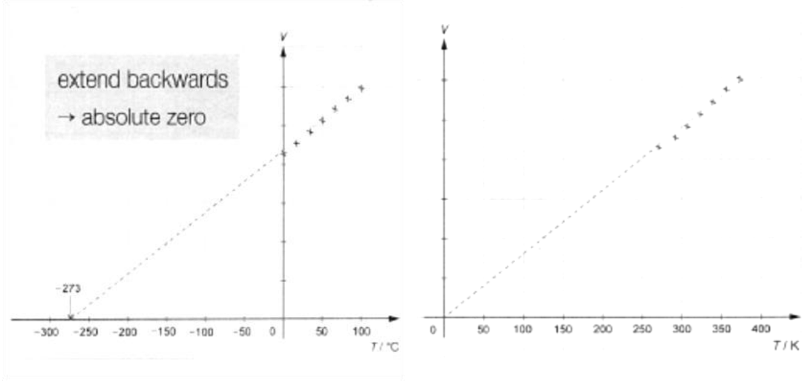
Charles' law

For a gas with a fixed mass (m) and pressure (P), its volume (V) is **directly proportional** to its **Kelvin temperature** (T)

Unit: (K)

$\boldsymbol{V \propto T}$ (constant P)

$\frac{V}{T}$ = constant or $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ (constant P)



13. If the volume of a balloon is 20 cm^3 at 30°C, what is the new volume if the temperature is changed to 20°C and the pressure is kept constant?

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{20}{30+273} = \frac{V_2}{20+273}$$

$$V_2 = 19.34 \text{ cm}^3$$

14. A gas of fixed mass and pressure is initially at temperature 156°C. When its volume is halved, find its temperature.

By Charles' law, $\frac{V}{T}$ = constant

When V is halved, T is halved.

Final $T = (156 + 273) \times \frac{1}{2} = 215$ K

$= (215 - 273)°\text{C} = -59°\text{C}$

15. A fixed mass of gas has its temperature changed from 127°C to 27°C at constant pressure. The ratio of the new volume to the old volume is

A. 27 : 127

B. 127 : 27

C. 3 : 4

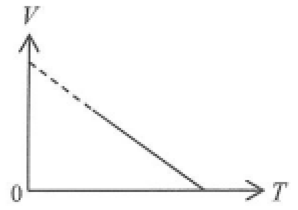
D. 4 : 3

By $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $\therefore \frac{V_1}{(127+273)} = \frac{V_2}{(27+273)}$ $\therefore \frac{V_2}{V_1} = \frac{3}{4}$

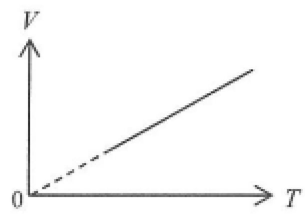
C

16. Which of the following graphs correctly shows the variation of volume V with absolute temperature T of a fixed mass of gas at constant pressure?

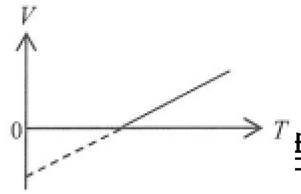
A.



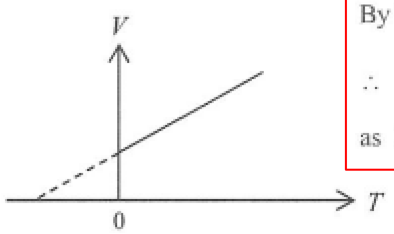
B.



C.



D.



B

By Volume-Temperature relation : $\frac{V}{T}$ = constant

$\therefore$ $V \propto T$

as T is the absolute temperature in Kelvin scale

General gas law

The relationship can be combined into a single expression

$$PV \propto T$$

$$\frac{PV}{T} = \text{constant or } \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

17. A weather balloon contains 4 m^3 of helium at the normal atmospheric pressure of 100 kPa and a temperature of 27 °C. What will be its volume when it rises to an altitude where the pressure is 80 kPa and the temperature is 7 °C?

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{(100000)(4)}{27+273} = \frac{(80000)V_2}{7+273}$$

$$V_2 = 4.67m^3$$

18. The pressure p of a fixed mass of ideal gas varies with the reciprocal of its volume V as shown. A student reads the graph and suggests that the temperature of the gas is not constant.

(i) Without calculation, explain that the student is correct. (1 mark)

(ii) If the Kelvin temperature of the gas is T when its pressure is 40 kPa, what is the Kelvin temperature of the gas when its pressure is 100 kPa? (2 marks)

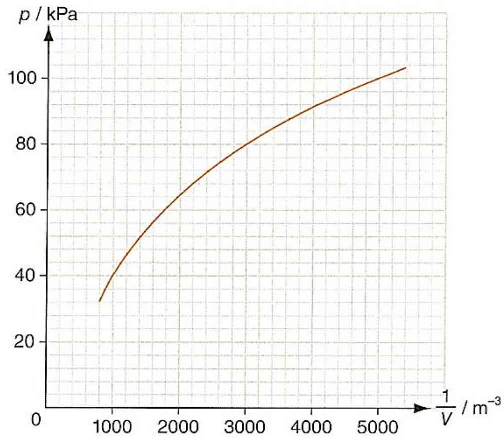
(a) (i) If the temperature is constant, p should be directly proportional to $\frac{1}{V}$, i.e. the graph should be a straight line passing through the origin. 1A

(ii) By $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$, 1M

$$T_2 = \frac{p_2}{p_1} \times \frac{V_2}{V_1} \times T_1$$

$$= \frac{100}{40} \times \frac{1000}{5000} \times T = 0.5T$$ 1A

∴ The temperature is $0.5T$.



The volume is directly proportional to the number of molecules N. As the number of molecules is represented by the number of mole.

- The unit of number of moles is **mole**
- One mole contains **6.02 × 10^{23}** molecules. This number is known as **Avogardro's number (N_A)**

General gas law can be represented by

PV = nRT

(The universal gas constant R is 8.31 J mol^{-1} K^{-1})

19. A car tyre has a constant volume of 12500 cm^3. The pressure of the air in the tyre is 275 kPa at a temperature of 30^{o}C. Assume air is an ideal gas. What is the amount of air in the tyre?

A. 1.2 mol

B. 1.4 mol

C. 1.6 mol

D. 1.8 mol

By $PV = nRT$

$\therefore\ (275 \times 10^3) \times (12500 \times 10^{-6}) = n \times (8.31) \times (30 + 273)$

$\therefore\ n = 1.4$ mol

B

20. A closed container of volume 1 m^3 contains an ideal gas. The temperature and pressure of the gas inside are 25^{o}C and 1.01 × 10^5 Pa respectively. Fins the number of gas molecules in the container.

A. 2.46×10^{25}

B. 2.93×10^{25}

C. 2.46×10^{26}

D. 2.93×10^{26}

By $PV = nRT$

$\therefore\ (1.01 \times 10^5)(1) = n(8.31)(25 + 273)$ $\qquad \therefore\ n = 40.8$ mol

Number of gas molecules $= 40.8 \times 6.02 \times 10^{23} = 2.46 \times 10^{25}$

A

21. A chamber contains an ideal gas at a pressure of 1.10 × 10^{-7} Pa and the temperature is 25^{o}C. Estimate the number of gas molecules per cubic metre in the chamber.

A. 2.67×10^{13}

B. 2.92×10^{13}

C. 3.19×10^{14}

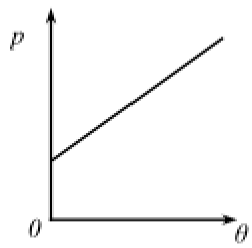
D. 3.94×10^{14}

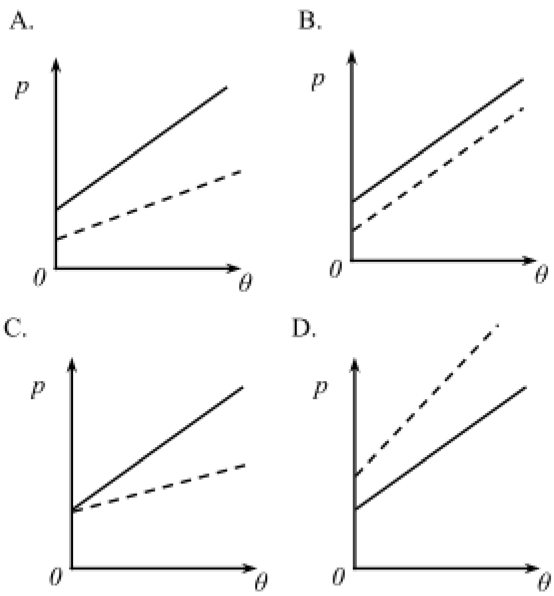
$$PV = nRT = \frac{N}{N_A}RT$$

$$\frac{N}{V} = \frac{PN_A}{RT} = \frac{(1.10 \times 10^{-7})(6.02 \times 10^{23})}{(8.31)(25 + 273)} = 2.67 \times 10^{13}\ \text{m}^{-3}$$

A

22. An ideal gas is contained in a closed vessel of fixed volume. The graph shows the variation of pressure p of the gas against its Celsius temperature θ. If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ.





3. Therefore, $pV = nR(\theta + 273)$ gives $p = \frac{nR}{V}\theta + \frac{273nR}{V}$ which is a straight line with slope $= \frac{nR}{V}$ and y-intercept $\frac{273nR}{V}$

4. Now, the number of gas molecules in the vessel is halved. Therefore, both the slope and y-intercept will decrease.

A

23. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.

D

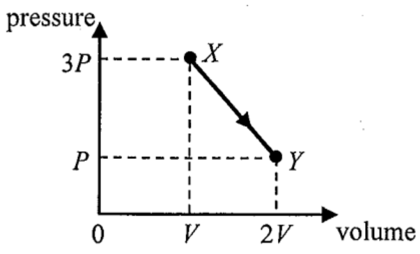
If the temperature of the gas at state Y is 25 °C, what is its temperature at state X?

A. −74.3 °C
B. 16.7 °C
C. 37.5 °C
D. 174 °C

$$\frac{(3P)(V)}{(T+273)} = \frac{(P)(2V)}{(25+273)}$$

$T = 174$

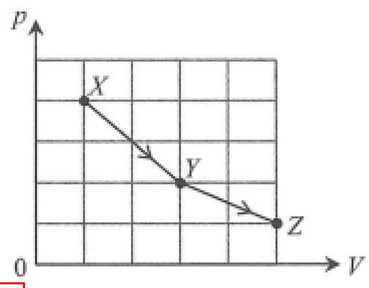
∴ Its temperature at state X is 174 °C.



24. An ideal gas of volume V and pressure p undergoes a change from state X to state Z via state Y along the path as shown in the p-V plot. Which of the following descriptions about the gas at X, Y and Z is correct?

A

A. The gas is at its coolest at X and is at its hottest at Y.
B. The gas is at its coolest at X and is at its hottest at Z.
C. The gas is at its coolest at Y and is at its hottest at X.
D. The gas has the same degree of hotness at X, Y and Z.



By $PV = nRT$ ∴ $T \propto PV$

∴ $T_X = 4k$ $T_Y = 6k$ $T_Z = 5k$

∴ The gas has highest temperature at Y and lowest temperature at X.

25. Two identical insulated cylinders X and Y are connected by a tube of negligible volume. Each cylinder has a volume of 2.0×10^{-2} m^3. The flow of gas between the cylinders is blocked by a tap on the tube. Cylinder X contains 1.2 mole of an ideal gas at 37 °C. Cylinder Y contains an ideal gas at pressure of 1.2×10^5 Pa and temperature of 37 °C.

(a) Find the number of mole of gas in cylinder Y. (2 marks)

(b) The tap is now opened and gas can flow freely between the cylinders. Using the fact that the total amount of gas is constant, determine the final pressure of the gas in the cylinders after equilibrium is attained. (2 marks)

Take $R = 8.31$ J mol^{-1} K^{-1}.

(a) By $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{1.2\times10^5 \times 2.0\times10^{-2}}{8.31\times(273+37)} = 0.932 \text{ mol}$$

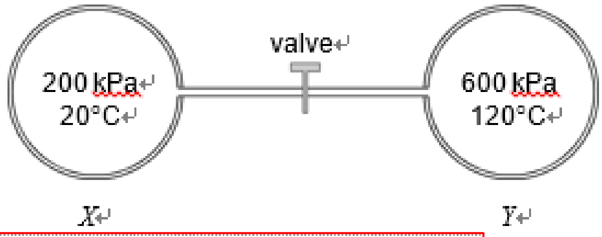
∴ There is 0.932 mol of gas in cylinder Y.

(b) Total number of mole of gas = 1.2 + 0.932 = 2.132

Final pressure of the gas

$$= \frac{nRT}{V} = \frac{2.132\times8.31\times(273+37)}{2\times2\times10^{-2}} = 1.37\times10^5 \text{ Pa}$$

26. X and Y are two identical insulated containers connected by tube with a valve which is initially closed. X contains an ideal gas at 200 kPa at 20°C. Y contains the same gas at 600 kPa at 120°C.
The valve is now opened. Suppose the temperatures of X and Y are kept unchanged, what is the final gas pressure when the system is steady?



Answer:

Let p be the final gas pressure.

Before the valve is opened, we have

$p_X V = n_X R T_X$ and $p_Y V = n_Y R T_Y$ (1M)

where V is the volume of each container.

The total number of moles of the gas is thus

$$n_X + n_Y = \frac{p_X V}{RT_X} + \frac{p_Y V}{RT_Y}$$
$$= \frac{V}{R}\left(\frac{200}{20+273} + \frac{600}{120+273}\right)$$
$$= 2.209\frac{V}{R}$$

After the valve is opened and the system becomes steady, we have

$pV = n'_X R T_X$ and $pV = n'_Y R T_Y$

The total number of moles of the gas is thus

$$n'_X + n'_Y = \frac{pV}{RT_X} + \frac{pV}{RT_Y}$$
$$= \frac{pV}{R}\left(\frac{1}{20+273} + \frac{1}{120+273}\right)$$
$$= 0.005957\frac{pV}{R}$$

Hence we have

$$n_X + n_Y = n'_X + n'_Y$$
$$2.209\frac{V}{R} = 0.005957\frac{pV}{R}$$
$$p = 371$$

The final gas pressure is 371 kPa.

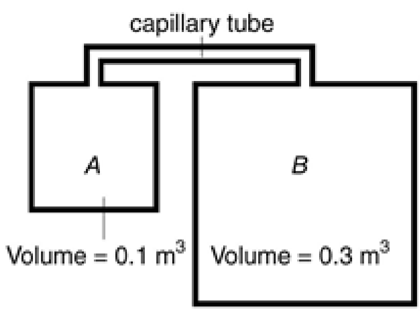
27. In the figure, two vessels A and B, of volume 0.1 m^3 and 0.3 m^3 respectively, are connected by a capillary tube. Both A and B contain nitrogen gas at 300 K and 100 kPa initially. A is then cooled down to 200 K while B is heated to 400 K. Take R = 8.31 J mol^{-1} K^{-1}.

(a) (i) Find the total number of moles of nitrogen in the vessels. (2 marks)

(ii) Find the final amount of nitrogen in each vessel. (3 marks)

(iii) How much nitrogen has passed through the capillary tube? (2 marks)

(iv) State two assumptions that you have made during calculations. (2 marks)



(a) (i) By $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{100\times1000\times(0.1+0.3)}{8.31\times300} = 16.0 \text{ mol}$$

∴ There are 16.0 mol of nitrogen.

(ii) Since the pressure in both vessels are the same,

$$\frac{n_A}{n_B} = \frac{V_A}{T_A}\times\frac{T_B}{V_B} = \frac{0.1}{200}\times\frac{400}{0.3} = \frac{2}{3}$$

$$\therefore n_A = 16.0\times\frac{2}{2+3} = 6.4 \text{ mol}$$

$$n_B = 16 - 6.4 = 9.6 \text{ mol}$$

∴ There are 6.4 mol in A and 9.6 mol in B.

(iii) Initial number of mole of gas in $A = 16.0\times\frac{0.1}{0.1+0.3} = 4.0$ mol

∴ Number of mole of gas pass through the tube = 6.4 – 4.0

= 2.4 mol

(iv) Nitrogen behaves as an ideal gas.

The capillary tube has a negligible volume.

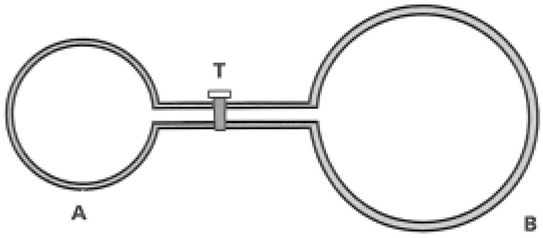
28. The containers A and B with volumes 100 cm^3 and 500 cm^3 respectively all jointed by a narrow tube of negligible volume as shown. Tap T is used to control gas flow. If it is initially closed. A contains an ideal gas at a pressure of 12 x 10^5 Pa. B is empty (vacuum). The temperature of the two containers is maintained at 0^{o}C by 2 separate water baths with pure melting ice. (R = 8.31 JK^{-1}mol^{-1})

(a) Calculate the number of mole of gas in container A

(b) The water bath for A is heated until water boils. Tap T remains opened. Hence, calculate

(i) the equilibrium pressure of the gas in the two containers

(ii) the net amount of gas in moles, that has passed through the connecting tube during the heating process



(a) By $pV = nRT$ *(1)*

$$n = \frac{12 \times 10^5 \times 100 \times 10^{-6}}{8.31 \times 273}$$ *(1)*

$$= \underline{0.0529}$$ *(1)*

(b) *(i)* Let number of moles in A and B at end be n_A, n_B

$n_A + n_B = 0.0529$ .... (1) *(1)*

and let equilibrium pressure be P

$P(100 \times 10^{-6}) = n_A(8.31)(373)$ .... (2) *(1)*

$P(500 \times 10^{-6}) = n_B(8.31)(273)$ .... (3) *(1)*

Solving,

$$P = \underline{2.094 \times 10^5 \text{ Pa}}$$ *(1)*

(ii) Before heating, number of moles in B

$$= 0.0529 \times \frac{500}{600}$$ *(1)*

$$= 0.044$$

After heating

$$n_B = \frac{pV}{RT} = 0.046$$ *(1)*

$\therefore$ net flows $= 0.046 - 0.044 = \underline{0.002 \text{ mole}}$ *(1)*